

REPORTS

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX_EMPLOYEES 100
```

```
#define MAX_BUDGETS 50
```

```
#define MAX_SUPPLIERS 100
```

```
#define MAX_ASSETS 100
```

```
//Budget Structure
```

```
int main() {
```

```
    char department[50];
```

```
    float allocatedBudget;
```

```
    float expenditure;
```

```
    float remainingBudget;
```

```
} Budget;
```

```
//Employee Structure
```

```
int main () {
```

```
    int employeeId;
```

```
    char name[50];
```

```
    char department[50];
```

```
    float basicSalary;
```

```
} Employee;
```

```
//Supplier Structure
```

```
int main () {  
    int supplierId;  
    char supplierName[50];  
    char email[100];  
    char phone[20];  
    char town[50];  
} Supplier;
```

//Asset Structure

```
int main () {  
    int assetId;  
    char assetName[50];  
    char assetType[50];  
    char purchaseDate[20];  
} Asset;
```

//Array and Counters

```
Employee employees[MAX_EMPLOYEES];  
Budget budgets[MAX_BUDGETS];  
Supplier suppliers[MAX_SUPPLIERS];  
Asset assets[MAX_ASSETS];  
int employeeCount = 0;  
int budgetCount = 0;  
int supplierCount = 0;  
int assetCount = 0;
```

//Add Employee

```
void addEmployee() {
```

```
if (employeeCount < MAX_EMPLOYEES) {  
    printf("\nEmployee storage is full.\n");  
    return;  
}  
printf("\n==== ADD EMPLOYEE =====\n");  
  
printf("Enter Employee ID: ");  
scanf("%d", &employees[employeeCount].employeeId);  
  
printf("Enter Employee Name: ");  
scanf(" %[^\\n]", employees[employeeCount].name);  
  
printf("Enter Department: ");  
scanf(" %[^\\n]", employees[employeeCount].department);  
  
printf("Enter Basic Salary: N$");  
scanf("%f", &employees[employeeCount].basicSalary);  
  
employeeCount++;  
  
printf("\nEmployee added successfully!\n");  
}  
  
void employeeReport(){  
    int i;  
  
    if (employeeCount == 0) {  
        printf("\nNo employees registered.\n");  
    }
```

```

    return;
}

float totalSalary = 0;

float averageSalary;

float highestSalary = employees[0].basicSalary;
float lowestSalary = employees[0].basicSalary;

for (i = 0; i < employeeCount; i++) {
    totalSalary += employees[i].basicSalary;

    if (employees[i].basicSalary > highestSalary){
        highestSalary = employees[i].basicSalary;
    }

    if (employees[i].basicSalary < lowestSalary){
        lowestSalary = employees[i].basicSalary;
    }
}

averageSalary = totalSalary / employeeCount;

printf("\n=====\\n");
printf("    EMPLOYEE REPORT\\n");
printf("=====\\n");

printf("Total Employees : %d\\n", employeeCount);
printf("Average Salary : N$%.2f\\n", averageSalary);
printf("Highest Salary : N$%.2f\\n", highestSalary);
printf("Lowest Salary : N$%.2f\\n", lowestSalary);

```

```

printf("\nEmployee Details\n");
printf("-----\n");

for (i = 0; i < employeeCount; i++) {
    printf("Employee ID: %d\n",
        employees[i].employeeId);

    printf("Name: %s\n",
        employees[i].name);

    printf("Department: %s\n",
        employees[i].department);

    printf("Basic Salary: N$%.2f\n",
        employees[i].basicSalary);

    printf("-----\n");
}
}

//Add Budget
void addBudget(){
    if (budgetCount >= MAX_BUDGETS) {
        printf("\nBudget storage is full.\n");
        return;
    }
    printf("\n===== ADD DEPARTMENT BUDGET =====\n");
}

```

```
printf("Enter Department: ");  
scanf("%s", budgets[budgetCount].department);  
  
printf("Enter Allocated Budget: N$");  
scanf("%f", &budgets[budgetCount].allocatedBudget);  
  
printf("Enter Expenditure: N$");  
scanf("%f", &budgets[budgetCount].expenditure);
```

```
// Calculate remaining budget  
budgets[budgetCount].remainingBudget =  
    budgets[budgetCount].allocatedBudget -  
    budgets[budgetCount].expenditure;  
budgetCount++;  
printf("\nBudget information added successfully!\n");  
}
```

```
void budgetReport() {  
    int i;  
    float totalAllocatedBudget = 0;  
    float totalExpenditure = 0;  
    float totalRemainingBudget;  
  
    if (budgetCount == 0){  
        printf("\nNo budget information registered.\n");  
        return;  
    }  
}
```

```
//Calculate total allocated budget
```

```

for (i = 0; i < budgetCount; i++){
    totalAllocatedBudget +=
        budgets[i].allocatedBudget;

    totalExpenditure +=
        budgets[i].expenditure;
}

totalRemainingBudget = totalAllocatedBudget - totalExpenditure;


printf("\n=====\\n");
printf("    BUDGET REPORT\\n");
printf("=====\\n");


printf("Total Allocated Budget : N$%.2f\\n",
    totalAllocatedBudget);


printf("Total Expenditure    : N$%.2f\\n",
    totalExpenditure);


printf("Remaining Budget    : N$%.2f\\n",
    totalRemainingBudget);


printf("\\nDepartment Budget Details\\n");
printf("-----\\n");


for (i = 0; i < budgetCount; i++) {
    printf("Department    : %s\\n",
        budgets[i].department);

```

```
printf("Allocated Budget : N$%.2f\n",  
    budgets[i].allocatedBudget);
```

```
printf("Expenditure    : N$%.2f\n",  
    budgets[i].expenditure);
```

```
printf("Remaining Budget : N$%.2f\n",  
    budgets[i].remainingBudget);
```

```
//Check if department exceeded budget
```

```
if (budgets[i].expenditure >  
    budgets[i].allocatedBudget) {  
    printf("STATUS : EXCEEDED BUDGET\n");  
}  
else{  
    printf("STATUS : Within Budget\n");  
}  
printf ("-----n");  
}  
}
```

```
//Add Supplier
```

```
void addSupplier(){  
if (supplierCount >= MAX_SUPPLIERS) {  
    printf("\nSupplier storage is full.\n");  
    return;  
}  
}
```

```
printf("\n===== ADD SUPPLIER =====\n");
```

```
printf("Enter Supplier ID: ");
```

```
scanf("%d", &suppliers[supplierCount].supplierId);
```

```
printf("Enter Supplier Name: ");
```

```
scanf(" %[^\\n]",
```

```
suppliers[supplierCount].supplierName);
```

```
printf("Enter Email: ");
```

```
scanf(" %[^\\n]",
```

```
suppliers[supplierCount].email);
```

```
printf("Enter Phone: ");
```

```
scanf(" %[^\\n]",
```

```
suppliers[supplierCount].phone);
```

```
printf("Enter Town: ");
```

```
scanf(" %[^\\n]",
```

```
suppliers[supplierCount].town);
```

```
supplierCount++;
```

```
printf("\nSupplier added successfully!\n");
```

```
}
```

```
void supplierReport(){
```

```
int i;
```

```

if (supplierCount == 0) {
    printf("\nNo suppliers registered.\n");
    return;
}

printf("\n===== \n");
printf("    SUPPLIER REPORT\n");
printf("===== \n");

printf("Total Suppliers: %d\n\n",
    supplierCount);

for (i = 0; i < supplierCount; i++){
    printf("Supplier ID   : %d\n",
        suppliers[i].supplierId);

    printf("Supplier Name : %s\n",
        suppliers[i].supplierName);

    printf("Email      : %s\n",
        suppliers[i].email);

    printf("Phone     : %s\n",
        suppliers[i].phone);

    printf("Town      %s\n",
        suppliers[i].town);

    printf("-----\n");

```

```

    }
}

//Add Asset
void addAsset(){
    if (assetCount >= MAX_ASSETS) {
        printf("\nAsset storage is full.\n");
        return;
    }
    printf("\n=== ADD MUNICIPAL ASSET ===\n");

    printf("Enter Asset ID: ");
    scanf("%d", &assets[assetCount].assetId);

    printf("Enter Asset Name: ");
    scanf(" %[^\\n]", assets[assetCount].assetName);

    printf("Enter Asset Type: ");
    scanf(" %[^\\n]", assets[assetCount].assetType);

    printf("Enter Purchase Value: N$");
    scanf("%f", &assets[assetCount].purchaseValue);
    assetCount++;

    printf("\nAsset added successfully!\n");
}

void assetReport(){

```

```

int i;

if (assetCount == 0){
    printf("\nNo municipal assets registered.\n");
    return;
}

printf("\n=====\\n");
printf("    ASSET REPORT\\n");
printf("=====\\n");

    printf("Total Assets: %d\\n\\n", assetCount);

for (i = 0; i < assetCount; i++) {

    printf("Asset ID: %d\\n", assets[i].assetId)
    printf("Asset Name: %s\\n", assets[i].assetName);
    printf("Asset Type: %s\\n", assets[i].assetType);
    printf("Purchase Value: N$%.2f\\n", assets[i].purchaseValue);
    printf("-----\\n");
}
}

//Main Menu

int main(){
    int choice;

    do {
        printf("\\n\\n");

        printf("=====\\n");

```

```
printf(" MUNICIPAL MANAGEMENT SYSTEM\n");
printf("=====\n");

printf("1. Add Employee\n");
printf("2. Add Department Budget\n");
printf("3. Add Supplier\n");
printf("4. Add Municipal Asset\n");

printf("\n--- REPORTS ---\n");

printf("5. Employee Report\n");
printf("6. Budget Report\n");
printf("7. Supplier Report\n");
printf("8. Asset Report\n");

printf("\n0. Exit\n");

printf("\nEnter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        addEmployee();
        break;

    case 2:
        addBudget();
        break;
```

case 3:

addSupplier();

break;

case 4:

addAsset();

break;

case 5:

employeeReport();

break;

case 6:

budgetReport();

break;

case 7:

supplierReport();

break;

case 8:

assetReport();

break;

case 0:

printf("\nThank you for using the system.\n");

break;

default:

```
printf("\nInvalid choice. Please try again.\n");
```

```
}
```

```
} while (choice != 0);
```

```
return 0;
```

```
}
```